

Adv C Module

1. Subjective

2.1 Basic Refreshers

1. What is the difference between overflow and underflow?
2. What are the basic data types in C? Explain their size and range.
3. What is the difference between signed and unsigned data types?
4. What are operators in C? Explain different types of operators.
5. What is type casting? Explain implicit and explicit type casting.
6. What is the difference between break and continue statements?

2.2 1D Pointers and Functions

1. What is a pointer? Why do we need pointers in C?
2. What is pointer arithmetic? Explain with examples.
3. What is the difference between array and pointer?
4. What is the difference between call by value and call by reference?
5. What is a null pointer and a dangling pointer?

2.3 String

1. How are strings represented?
2. What is the difference between character array and string literal?
3. What is the importance of the null character (\0) in strings?
4. What are common string handling functions in C?
5. What is the difference between strcpy() and strncpy()?

2.4 Storage classes and memory segments

1. What are storage classes in C? Explain auto, register, static, and extern.
2. What is the difference between global and local variables?
3. Explain the memory layout.
4. What is the difference between stack and heap memory?
5. What happens when stack overflow occurs?

2.5 2D Pointers and DMA

1. Explain memory segments.
2. What is a function pointer?
3. What is the difference between pointer to array and array of pointer.
4. What is dynamic memory allocation?
5. Explain malloc(), calloc(), realloc(), and free().
6. What is a double pointer? Why is it used?
7. What is the difference between static and dynamic memory allocation?
8. What is a memory leak and how can it be prevented?

2.6 Preprocessing

1. What is a preprocessor in C?
2. What is the purpose of #define?
3. What is conditional compilation?
4. What are header files and why are they used?
5. What are macros and what are their advantages and disadvantages?

2.7 UDT

1. What is the difference between structure and union?
2. What is a structure?
3. What are user-defined data types?
4. What is an enumeration (enum)?
5. What is typedef and why is it used?
6. What is the difference between structure padding and alignment?

2.8 Miscellaneous

1. Explain static and volatile with sample code.
2. What is the volatile keyword and when is it used?
3. What is the const keyword?

2.9 FILE I/O

1. What is file handling?
2. What are the different file opening modes?
3. What is the difference between text files and binary files?
4. What is the difference between fprintf/fscanf and fread/fwrite?
5. What is file pointer (FILE *) and how does it work?

2.10 Projects

1. Explain your Address Book project.
2. How did you implement file handling in the Address Book project?
3. Why did you use a CSV file?
4. How do you load data into the program at the beginning?
5. Explain the Steganography project.
6. Why is it called LSB image steganography?
7. Explain the byte-to-LSB logic with code.
8. How do you use bitwise operators in steganography?
9. If only the encoder and decoder know the magic string, how can a third person find it?

2. Programming

1. WAP to swap two numbers without using a temporary variable.
2. WAP to swap two numbers using pointers.
3. WAP to reverse a string using a loop and recursion.
4. WAP to reverse words in a string.
5. WAP to reverse a string such that the first character of each word is capital after reversing.
6. WAP to convert integer to ASCII.
7. WAP to convert binary to decimal.
8. WAP to convert octal to decimal.
9. WAP to convert hex to decimal.
10. WAP to convert string to integer.
11. WAP to count set bits.
12. WAP to check whether a string is palindrome.
13. WAP to check whether binary representation of an integer is palindrome.
14. WAP to count vowels in a string.
15. WAP to get string length without using strlen().

16. WAP to concatenate strings without using strcat().
17. WAP to allocate memory using malloc for 5 integers.
18. WAP to print the 3rd character using a character pointer.
19. WAP to find the second largest number in an array.
20. WAP to find missing elements in an array.
21. WAP to find duplicate elements in an array.
22. WAP to remove duplicates from sorted arrays.
23. WAP to sort an array.
24. WAP to print alternate elements in an array.
25. WAP to clear n bits.
26. WAP to swap particular bits.
27. WAP to swap nibbles.
28. WAP to set all even bits to 0.
29. WAP for pyramid pattern.
30. Write structure to read student details and print the highest score first.
31. WAP to read from one file and write to another file.
32. WAP to get file size.
33. WAP to read data from one file, modify it, and write into another file.
34. WAP to merge two arrays, make unique elements, and find the median.
35. WAP to combine two 16-bit integers into one 32-bit integer (MSB & LSB logic).
36. WAP to toggle bits from 13th to 26th position.
37. WAP to print bit positions of 1s from 13th to 21st bit.
38. WAP to count substrings with equal consecutive 0s and 1s.
39. WAP for GCD.
40. WAP to print an array of size N with duplicate elements N/2 without temp variable.
41. Solve the Chessboard Coloring problem using a 2D array approach and compute the result modulo $10^9 + 7$.
42. WAP to load boxes into trucks without exceeding the capacity of each truck.
43. WAP to convert a string into a neat string such that each character is either fully lowercase or uppercase with minimum cost.

44. WAP to build the longest palindrome from a given string and return its length.
45. WAP to check if two strings are anagrams using dynamic memory and store the result in a file.
46. WAP to check if a string can be obtained by cyclic rotation.
47. WAP to find mean and variance of matrix and plot graph.
48. WAP to extract bits from 13th to 21st position and print positions of 1's and count them.

MC Module

1. Basic electronics

1. What is the difference between gates and flip-flops?
2. Truth table for AND, OR, NAND, NOR, XOR, XNOR.
3. Which gates are universal gates and why?
4. WAP to implement XOR using NOR gate.
5. What is channel capacity?
6. What are filters? Explain low pass and band pass filters.
7. What is a rectifier? Explain half-wave and full-wave rectifiers.
8. What are the types of transistors you have learned?
9. Explain the types of BJT. (Draw the diagrams.)

2. Basics

1. How does printf work in a microcontroller?
2. How to run an application in a microcontroller?
3. How does the microcontroller know the peripheral is initialized?
4. What is the difference between a microprocessor and a microcontroller?
5. What is the difference between an 8-bit and a 32-bit microcontroller?
6. What is the difference between Von Neumann and Harvard architecture?
7. How to dump code into a microcontroller?
8. How is CLCD connected to EEPROM?

3. Interrupts

1. Explain timers in detail.
2. How does the timer work internally?
3. What is an interrupt in a microcontroller?
4. Why are interrupts used instead of polling?
5. What are the different types of interrupts?
6. What is an Interrupt Service Routine (ISR)?

4. Projects

1. Explain the Car Black Box project.
2. Why do we need to store gear information if speed is already stored?
3. Explain the architecture.
4. How did you interface the CLCD?
5. How did you interface the digital keypad?
6. How can this be implemented in a real car?
7. How do you connect it with gears?

5. ADC

1. What is quantization?
2. What is the resolution?
3. What is ADC?
4. If voltage is 5V, how many steps?
5. If transmitting RF signals, is ADC required?

6. Embedded Systems

1. What is an embedded system? How is it different from a general-purpose system?
2. What are the main components of an embedded system?
3. What is the difference between a microcontroller and a microprocessor?
4. What are real-time systems? Explain hard and soft real-time systems.
5. What is an interrupt? How does it work in a microcontroller?

7. Protocols

1. Explain I2C protocol.
2. What is modulation and its types of modulation?
3. Explain sampling in detail.
4. What is the difference between SPI, I2C, CAN.
5. Explain UART protocol.
6. Why are pull-up resistors used in I²C instead of pull-down resistors?

8. PWM

1. WAP to control servo motors using PWM.
2. What is PWM and why is it used in embedded systems?
3. What is the duty cycle in PWM? How is it calculated?
4. What is the difference between PWM and analog voltage control?
5. How is PWM generated in a microcontroller?
6. What are the applications of PWM in real-time systems?

9. Programming

1. Write a program to generate a 1 ms delay using a timer.
2. Write a program to toggle an LED using a GPIO pin.
3. Write a program to generate PWM with 50% duty cycle.
4. Write a program to read analog input using ADC.
5. Write a program to toggle LEDs inside ISR.

CPP Module

1. What is the difference between C and C++?
2. What is a class and an object?
3. Explain the concept of encapsulation with an example.
4. What is inheritance? Explain different types of inheritance.
5. What is polymorphism? Explain compile-time and run-time polymorphism.
6. What is a virtual function? Why is it needed?
7. What is a constructor and destructor? When are they called?
8. What is the difference between structure and class in C++?
9. What is a friend function? When should it be used?
10. What is the difference between a pointer and a reference in C++?
11. What is the difference between structure and class?

Ds Module

1. Basics

1. What is a data structure? Why do we need data structures?
2. What is the difference between linear and non-linear data structures?
3. What is time complexity and space complexity?
4. What is the difference between static and dynamic data structures?
5. What is Abstract Data Type (ADT)?

2. Linked lists

1. WAP to find a loop in the linked list.
2. WAP to insert at last in a single linked list.
3. WAP to find the middle element of a linked list without traversing twice.
4. WAP to add elements in the linked list.
5. What is a linked list? How is it different from an array?
6. Explain types of linked lists (Singly, Doubly, Circular).

3. Stack

1. What is a stack? Explain its basic operations.
2. What is the difference between stack and queue?
3. How can you implement a stack using an array?
4. How can you implement a stack using a linked list?
5. What are real-time applications of stack?

4. Searching and Sorting Techniques

1. What is the difference between linear search and binary search?
2. What is the time complexity of binary search?
3. Explain bubble sort and its time complexity.
4. What is the difference between merge sort and quick sort?
5. What is a stable sorting algorithm?

5. Queue

1. What is a queue? Explain its basic operations.
2. What is the difference between a linear queue and circular queue?
3. How can you implement a queue using an array?
4. How can you implement a queue using a linked list?
5. What are applications of queue in real-world systems?

6. Hashing

1. What is hashing? Why is it used?
2. What is a hash function? What are its properties?
3. What is a collision in hashing?
4. What are collision resolution techniques?
5. What is the difference between open addressing and chaining?

7. Trees

1. What is a tree data structure?
2. What is the difference between binary tree and binary search tree?
3. What are tree traversals? Explain inorder, preorder, and postorder.
4. What is the height and depth of a tree?
5. What is a balanced binary tree?

8. Projects

1. Explain your APC project.
2. Write the logic to add two numbers using a linked list.
3. What are the use cases of the APC project?
4. What is the purpose of the APC project?
5. Explain the Inverted Search project in detail.

LI Module

1. Basics

1. What is OS?
2. How do you know how many bytes are allocated for kernel space?
3. How to run applications in kernel space?
4. How to initialize an OS?
5. Explain ls, top, grep, who commands.

2. System call

1. Which is more efficient: standard libraries or system calls?
2. What is a system call? Why is it needed?
3. What is the difference between system calls and library functions?
4. What happens internally when a system call is invoked?
5. What are the major categories of system calls?
6. What is the difference between user mode and kernel mode?

3. Networking

1. What is the OSI model? Explain its layers.
2. What is the difference between TCP and UDP?
3. What is IP addressing? Explain IPv4 and IPv6.
4. What is subnetting? Why is it used?
5. What is DNS and how does it work?

4. Process

1. What is a process?
2. What is the difference between a process and a program?
3. What are the different states of a process?
4. What is a Process Control Block (PCB)?
5. What is context switching?

5. IPC

1. Explain Linux pipes and FIFO.
2. What is IPC and why is it required?
3. What are the different IPC mechanisms in Linux?
4. What is the difference between pipes and message queues?
5. What is shared memory and how does it work?

6. Signal

1. What is SIGSEGV?
2. What is a signal in Unix/Linux?
3. What is the difference between SIGINT and SIGKILL?
4. What is a signal handler?
5. What is the difference between synchronous and asynchronous signals?
6. How does a process handle signals?

7. Socket

1. What is a socket in networking?
2. What is the difference between TCP socket and UDP socket?
3. What are the steps involved in TCP client-server communication?
4. What is bind(), listen(), accept() used for?
5. What is the difference between blocking and non-blocking sockets?

8. Threads

1. What is a thread? How is it different from a process?
2. What are the advantages of multithreading?
3. What is race condition?
4. What is the difference between mutex and semaphore in threads?
5. What is thread synchronization?

9. Process and Memory Management

1. What is virtual memory?
2. What is paging?
3. What is segmentation?
4. What is demand paging?
5. What is memory fragmentation? Explain internal and external fragmentation.

General Questions

Aptitude

1. Why do you want to join the company?
2. What is your goal?
3. What do you expect from this role?
4. Why this company?
5. Explain your prior work experience.
6. If you are assigned a new module that you don't know and asked to complete it within 2 days, how will you approach it?
7. If your presence is required both at your office and for an important family matter, which one will you prefer?
8. Explain the Vending Machine project approach.
9. How would you test a gallery application with multiple test case scenarios?
10. Solve the Submarine Challenge problem using bit manipulation techniques.
11. Determine the lexicographical order of strings using an optimized time-complexity approach.
12. Solve the Gym Exercise Completion Scheduling problem using an efficient scheduling algorithm.
13. Solve the Keypad Sequence Matching problem using an appropriate string or pattern-matching technique.
14. Count the number of inverted right-angled triangles present in a given matrix.
15. What is testing in software engineering?
16. What is validation?
17. What are the different types of software testing?
18. What is black box testing?
19. What is white box testing?
20. What is regression testing?
21. What is unit testing?
22. What is integration testing?
23. What is system testing?
24. What is a test case?
25. Explain scenario-based testing with an example.

26. What projects have you done?
27. Which concepts did you use?
28. Which debugger did you use?
29. Discuss networking protocols you know.
30. What is your salary expectation?
31. What is your CTC expectation?
32. What do you know about the company?
33. Where do you see yourself in 2 years?
34. Will you relocate to Bangalore if hired?
35. What frequency bands are used in Bluetooth and Wi-Fi modules?